MANIPAL UNIVERSITY JAIPUR

School of Computer Science & Engineering

Department of Computer Science & Engineering

PROJECT SYNOPSIS

for Android App Development

Project Title

“Pocketly: Personal Finance & Expense Tracker”

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Course / Lab	Android App Development Laboratory
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SYNOPSIS

1. Project Title

Pocketly: Personal Finance & Expense Tracker

2. Abstract

This project proposes the design and development of an Android-based mobile application for managing personal finances, daily transaction histories, connected bank accounts, and spending analytics. The application will demonstrate the practical application of Android app development concepts, including user interface design, activity and screen navigation, event handling, input validation, data storage, and application lifecycle management. The project will be developed using Android Studio with Kotlin and XML where appropriate. The proposed application is intended to provide an organized, user-friendly, and efficient solution for performing common tasks related to the selected problem domain. The project will emphasize responsive user interaction, proper data handling, usability, and reliable application behavior. Through this implementation, students will gain hands-on experience in converting a real-world problem into a functional Android mobile application.

3. Problem Statement

The objective of this project is to identify a real-world problem that can be addressed through a mobile application and develop an Android-based solution that efficiently allows users to enter, process, view, update, and manage the required information. The application should provide an intuitive user interface, validate user input, handle user interactions correctly, and maintain data consistently throughout the application.

4. Objectives

· To analyze the selected real-world problem and identify its functional requirements.

· To design an appropriate and user-friendly Android application interface.

· To implement application screens using XML layouts and Kotlin.

· To handle user interactions using buttons, input fields, RadioButtons, CheckBoxes, and other UI components as required.

· To implement navigation between activities/screens.

· To validate user input and provide appropriate feedback to the user.

· To implement suitable local data storage and retrieval where required.

· To demonstrate Android application lifecycle and event-handling concepts.

· To test the application for functionality, usability, and basic error handling.

5. Proposed Application Components

The project may contain multiple application screens and components such as Dashboard Feed (HomeScreen with Balance & Spending cards), Transaction Management (TransactionsScreen with Search & Filters), and Analytics & Profile (AnalyticsScreen & ProfileScreen). The exact screens, features, UI components, and data requirements will be finalized after analysis of the selected problem domain.

6. Android Development Concepts to be Demonstrated

· Android Studio project structure and configuration

· Kotlin programming for Android application logic

· XML layouts and user interface components

· Activities and activity lifecycle

· Intent-based navigation between screens

· Event handling for buttons and other UI controls

· EditText, TextView, Button, ImageView, RadioButton, CheckBox, Spinner, and other suitable components

· Input validation and user feedback using Toasts or dialogs

· RecyclerView and Adapter where applicable

· Local data storage using SQLite/Room/SharedPreferences, where applicable

· Basic exception handling and application testing

7. Expected Outcome

The expected outcome is a functional Android mobile application prototype capable of addressing the selected problem domain. The project will demonstrate appropriate Android UI design, Kotlin-based application logic, event handling, screen navigation, input validation, data management where required, and practical application of Android app development concepts.

8. Student Details

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Section / Semester	**Section I / 5th Semester**
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